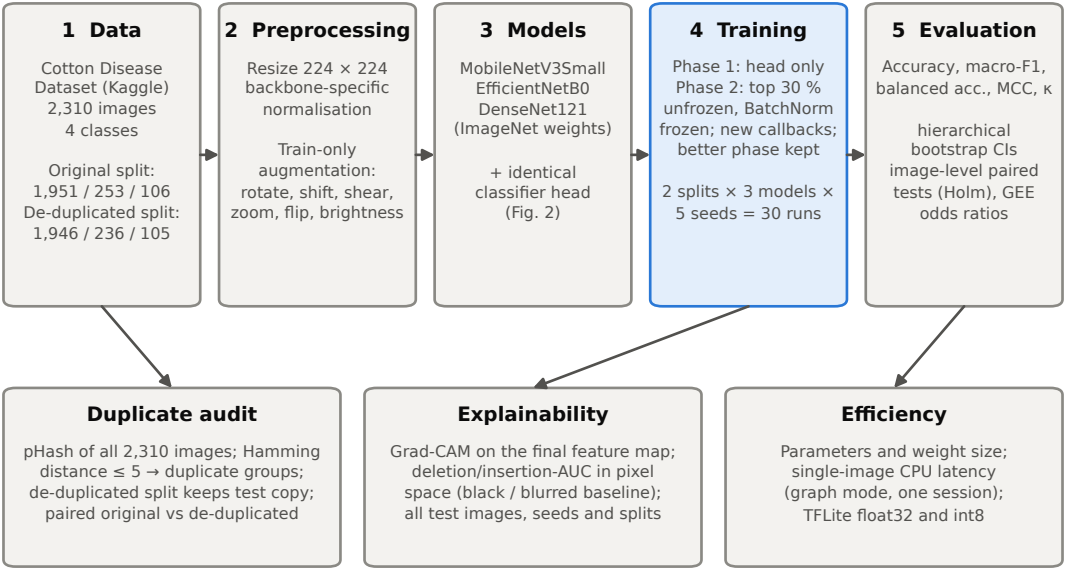

1 Data
Cotton Disease Dataset (Kaggle)
2,310 images
4 classes
Original split:
1,951 / 253 / 106
De-duplicated split:
1,946 / 236 / 105
2 Preprocessing
Resize 224 × 224
backbone-specific normalisation
Train-only augmentation:
rotate, shift, shear, zoom, flip, brightness
3 Models
MobileNetV3Small
EfficientNetB0
DenseNet121
(ImageNet weights)
+ identical classifier head (Fig. 2)
4 Training
Phase 1: head only
Phase 2: top 30 % unfrozen, BatchNorm frozen; new callbacks; better phase kept
2 splits × 3 models × 5 seeds = 30 runs
5 Evaluation
Accuracy, macro-F1, balanced acc., MCC, κ
hierarchical bootstrap CIs
image-level paired tests (Holm), GEE odds ratios
Duplicate audit
pHash of all 2,310 images; Hamming distance ≤ 5 → duplicate groups; de-duplicated split keeps test copy; paired original vs de-duplicated
Explainability
Grad-CAM on the final feature map; deletion/insertion-AUC in pixel space (black / blurred baseline); all test images, seeds and splits
Efficiency
Parameters and weight size; single-image CPU latency (graph mode, one session); TFLite float32 and int8